

arduinoBlue

Arduino C++ Notes

Complete Programming Guide

Beginner se Advanced tak

Is notebook mein Arduino ke liye C++ ke sabhi important concepts aur examples hain.

Topics: Digital I/O • Analog • Serial •
Sensors • Motors • Interrupts • Timing

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1 Arduino Basics — Shuruat

1.1 Arduino Program ki Structure

Har Arduino program mein **do main functions** hote hain:

arduinoBlue!20 Function	Kab Chalta Hai	Kaam
setup()	Sirf ek baar — start par	Sab kuch initialize karna
loop()	Baar baar — hamesha	Main kaam karna

Ek Basic Arduino Program

```

1 void setup() {
2     // Yeh sirf ek baar chalega jab Arduino start hoga
3     pinMode(LED_BUILTIN, OUTPUT); // Pin 13 ko output banao
4 }
5
6 void loop() {
7     // Yeh baar baar chalta rahega hamesha
8     digitalWrite(LED_BUILTIN, HIGH); // LED ON
9     delay(1000);                      // 1 second ruko
10    digitalWrite(LED_BUILTIN, LOW);  // LED OFF
11    delay(1000);                      // 1 second ruko
12 }
```

1.1.1 Keywords ka Matlab

- **void** — Function kuch return nahi karta (khali haath wapas)
- **setup** — Function ka naam (Arduino ka special naam)
- **()** — Function ka darwaza, andar parameters jaate hain
- **{}** — Function ka body, andar sab kaam likhte hain
- **;** — Har statement ke end mein lagta hai (zaroori!)

Yaad Rakho (Remember)

setup() aur loop() Arduino mein **reserved names** hain. Inhe change mat karo — Arduino inhe automatically dhundta hai!

2 Digital Input / Output

2.1 LED Blink — Simple On/Off

2.1.1 Concept

- **pinMode(pin, mode)** — Pin ko INPUT ya OUTPUT set karo
- **digitalWrite(pin, value)** — Pin ko HIGH (5V) ya LOW (0V) karo
- **delay(ms)** — Milliseconds tak ruko

Example 1: LED Blink

```
1 // Pin 13 par LED lagaao
2 int ledPin = 13; // Variable – pin number store karta hai
3
4 void setup() {
5     pinMode(ledPin, OUTPUT); // ledPin ko output banao
6 }
7
8 void loop() {
9     digitalWrite(ledPin, HIGH); // LED ON karo
10    delay(500); // 0.5 second ruko
11    digitalWrite(ledPin, LOW); // LED OFF karo
12    delay(500); // 0.5 second ruko
13    // Yeh loop repeat hoga – LED blink karega
14 }
```

2.2 Multiple LEDs

Example 2: Teen LED ka Chase Effect

```
1 // Teen alag pins par teen LED
2 int led1 = 11;
3 int led2 = 12;
4 int led3 = 13;
5
6 void setup() {
7     pinMode(led1, OUTPUT);
8     pinMode(led2, OUTPUT);
9     pinMode(led3, OUTPUT);
10 }
11
12 void loop() {
13     // Ek ek karke LED on karo – chase effect
14     digitalWrite(led1, HIGH);
15     delay(200);
16     digitalWrite(led1, LOW);
17 }
```

```
18 digitalWrite(led2, HIGH);
19 delay(200);
20 digitalWrite(led2, LOW);
21
22 digitalWrite(led3, HIGH);
23 delay(200);
24 digitalWrite(led3, LOW);
25 }
```

2.3 Button se LED Control

2.3.1 digitalRead — Button Press Padhna

Example 3: Button se LED On/Off

```
1 int buttonPin = 2; // Button pin 2 par
2 int ledPin    = 13; // LED pin 13 par
3 int buttonState; // Variable – button ki state store karta
  hai
4
5 void setup() {
6   pinMode(buttonPin, INPUT); // Button = INPUT
7   pinMode(ledPin, OUTPUT);   // LED = OUTPUT
8 }
9
10 void loop() {
11   buttonState = digitalRead(buttonPin); // Button opadh
12
13   if (buttonState == HIGH) {           // Agar button dabaya
14     digitalWrite(ledPin, HIGH);        // LED ON
15   } else {                             // Warna
16     digitalWrite(ledPin, LOW);         // LED OFF
17   }
18 }
```

Tip

INPUT_PULLUP use karo agar external resistor nahi hai:
`pinMode(buttonPin, INPUT_PULLUP);`
Is case mein logic ulta hoga — button dabane par LOW aayega.

Example 4: Toggle Button (Press karo ON — Phir Press karo OFF)

```
1 int buttonPin = 2;
2 int ledPin    = 13;
3 bool ledState = false; // LED abhi OFF hai
```

```
4 bool lastButton = false; // Pehle button ka state
5
6 void setup() {
7     pinMode(buttonPin, INPUT_PULLUP);
8     pinMode(ledPin, OUTPUT);
9 }
10
11 void loop() {
12     bool currentButton = (digitalRead(buttonPin) == LOW); //
        Dabaya?
13
14     // Sirf tab toggle karo jab nayi press ho
15     if (currentButton == true && lastButton == false) {
16         ledState = !ledState; // ON hoga toh OFF, OFF
            hoga toh ON
17         digitalWrite(ledPin, ledState); // LED update karo
18     }
19
20     lastButton = currentButton;
21     delay(50); // Debounce – bounce se bachao
22 }
```


3 Analog Input / Output

3.1 Analog Read — Sensor Value Padhna

Arduino mein **Analog pins (A0-A5)** 0 se 1023 tak value dete hain.

arduinoBlue!20 Voltage	analogRead() Value
0V	0
2.5V	512
5V	1023

Example 5: Potentiometer (Variable Resistor) se LED Control

```

1 int potPin = A0;    // Potentiometer A0 par lagao
2 int ledPin = 9;     // LED PWM pin par (3,5,6,9,10,11)
3 int potValue;       // Potentiometer ki value (0-1023)
4 int brightness;     // LED ki brightness (0-255)
5
6 void setup() {
7     pinMode(ledPin, OUTPUT);
8     Serial.begin(9600); // Serial monitor shuru karo
9 }
10
11 void loop() {
12     potValue = analogRead(potPin);    // 0 se 1023 tak opadh
13     brightness = potValue / 4;        // 0-1023 ko 0-255 mein
14     // badlo
15     analogWrite(ledPin, brightness);  // LED ki brightness set
16     // karo
17
18     Serial.print("Pot Value: ");
19     Serial.print(potValue);
20     Serial.print(" | Brightness: ");
21     Serial.println(brightness);
22
23     delay(100);
24 }

```

Yaad Rakho (Remember)

analogWrite() sirf PWM pins par kaam karta hai:
 Arduino Uno mein: Pin **3, 5, 6, 9, 10, 11** (tilde ~ sign se pehchano)

3.2 map() Function — Value Range Badalna

Example 6: map() se Temperature Display

```
1 int tempPin = A1;  // Temperature sensor A1 par
2
3 void setup() {
4     Serial.begin(9600);
5 }
6
7 void loop() {
8     int rawValue = analogRead(tempPin); // 0-1023
9
10    // map(value, fromLow, fromHigh, toLow, toHigh)
11    // 0-1023 ko 0-100 degree mein badlo
12    int temperature = map(rawValue, 0, 1023, 0, 100);
13
14    Serial.print("Temperature: ");
15    Serial.print(temperature);
16    Serial.println(" C");
17
18    delay(500);
19 }
```

4 Serial Communication

4.1 Serial Monitor se Baat Karna

Example 7: Serial Monitor par Data Print Karna

```
1 int sensorPin = A0;
2
3 void setup() {
4     Serial.begin(9600); // 9600 baud rate par serial shuru karo
5     Serial.println("Arduino Ready!"); // Ek baar print karo
6 }
7
8 void loop() {
9     int value = analogRead(sensorPin);
10
11     Serial.print("Value = "); // Bina newline ke print karo
12     Serial.println(value);    // Value print karo + newline
13
14     delay(1000);
15 }
```

4.2 Serial se Input Lena

Example 8: Computer se LED Control karo

```
1 int ledPin = 13;
2 char command; // Ek character store karega
3
4 void setup() {
5     pinMode(ledPin, OUTPUT);
6     Serial.begin(9600);
7     Serial.println("Commands: 'H' = ON, 'L' = OFF");
8 }
9
10 void loop() {
11     if (Serial.available() > 0) { // Koi data aaya hai?
12         command = Serial.read(); // Ek character opadh
13
14         if (command == 'H') {
15             digitalWrite(ledPin, HIGH);
16             Serial.println("LED ON!");
17         }
18         else if (command == 'L') {
19             digitalWrite(ledPin, LOW);
20             Serial.println("LED OFF!");
21         }
22         else {
```

```
23     Serial.println("Invalid command!");  
24 }  
25 }  
26 }
```

5 Variables aur Data Types

5.1 C++ mein Data Types

arduinoBlue!20 Type	Size	Range	Example
int	2 bytes	-32768 to 32767	int x = 10;
long	4 bytes	-2B to 2B	long t = 100000;
float	4 bytes	Decimal numbers	float v = 3.14;
bool	1 bit	true / false	bool on = true;
char	1 byte	One character	char c = 'A';
String	Variable	Text	String s = "Hello";
byte	1 byte	0 to 255	byte b = 255;

Example 9: Variables ka Use

```

1 int    counter = 0;           // Poora number – counter ke liye
2 float  voltage = 0.0;        // Decimal number – voltage ke liye
3 bool   isOn    = false;      // Sirf true ya false
4 char   grade   = 'A';        // Ek akshar
5 String name    = "LED";      // Text
6 long   timer   = 0;          // Bada number – time ke liye
7
8 void setup() {
9     Serial.begin(9600);
10 }
11
12 void loop() {
13     counter++; // counter = counter + 1 (shortcut)
14
15     int raw = analogRead(A0);
16     voltage = raw * (5.0 / 1023.0); // 0-1023 ko voltage mein
17                                     badlo
18     Serial.print("Count: "); Serial.println(counter);
19     Serial.print("Voltage: "); Serial.println(voltage);
20
21     delay(1000);
22 }

```

5.2 Arrays — Kai Values Ek Saath

Example 10: Array se Multiple LEDs Control

```

1 // Array – kai values ek group mein
2 int ledPins[] = {8, 9, 10, 11, 12, 13}; // 6 LEDs ke pins
3 int numLeds   = 6;                       // Kitne LEDs hain
4

```

```
5 void setup() {  
6   // for loop se sabhi pins setup karo  
7   for (int i = 0; i < numLeds; i++) {  
8     pinMode(ledPins[i], OUTPUT); // ledPins[0], [1], [2]...  
9   }  
10 }  
11  
12 void loop() {  
13   // Ek ek karke ON karo  
14   for (int i = 0; i < numLeds; i++) {  
15     digitalWrite(ledPins[i], HIGH);  
16     delay(100);  
17   }  
18  
19   // Ek ek karke OFF karo  
20   for (int i = numLeds - 1; i >= 0; i--) {  
21     digitalWrite(ledPins[i], LOW);  
22     delay(100);  
23   }  
24 }
```

6 Control Structures – Decision aur Loops

6.1 if / else if / else

Example 11: Light Sensor se Automatic LED

```
1 int ldrPin = A0; // LDR (Light sensor) A0 par
2 int ledPin = 13;
3 int ldrValue;
4
5 void setup() {
6     pinMode(ledPin, OUTPUT);
7     Serial.begin(9600);
8 }
9
10 void loop() {
11     ldrValue = analogRead(ldrPin); // 0 = andhera, 1023 = ujala
12
13     if (ldrValue < 300) {
14         // Bahut andhera
15         digitalWrite(ledPin, HIGH);
16         Serial.println("Dark - LED ON");
17     }
18     else if (ldrValue < 700) {
19         // Theek thak roshni
20         Serial.println("Normal light");
21     }
22     else {
23         // Bahut roshan
24         digitalWrite(ledPin, LOW);
25         Serial.println("Bright - LED OFF");
26     }
27
28     delay(500);
29 }
```

6.2 for Loop

Example 12: for Loop se LED Fade In/Out

```
1 int ledPin = 9; // PWM pin zaroori hai
2
3 void setup() {
4     pinMode(ledPin, OUTPUT);
5 }
6
7 void loop() {
8     // Fade IN: 0 se 255 tak brightness badhao
```

```
9   for (int brightness = 0; brightness <= 255; brightness++) {
10       analogWrite(ledPin, brightness);
11       delay(10); // Thoda ruko har step par
12   }
13
14   // Fade OUT: 255 se 0 tak brightness oghata
15   for (int brightness = 255; brightness >= 0; brightness--) {
16       analogWrite(ledPin, brightness);
17       delay(10);
18   }
19 }
```

6.3 while Loop

Example 13: while Loop — Button Dabne tak Wait Karo

```
1  int buttonPin = 2;
2  int ledPin    = 13;
3
4  void setup() {
5      pinMode(buttonPin, INPUT_PULLUP);
6      pinMode(ledPin, OUTPUT);
7      Serial.begin(9600);
8  }
9
10 void loop() {
11     Serial.println("Button dabao...");
12
13     // Jab tak button nahi daba, yahan ruko
14     while (digitalRead(buttonPin) == HIGH) {
15         // Kuch mat karo – bas wait karo
16         delay(10);
17     }
18
19     // Button dab gaya!
20     Serial.println("Button daba gaya!");
21     digitalWrite(ledPin, HIGH);
22     delay(1000);
23     digitalWrite(ledPin, LOW);
24 }
```


7 Functions Banana — Khud ka Function

7.1 Custom Function Kaise Banate Hain

Example 14: Custom Function — LED Blink Function

```
1 int ledPin = 13;
2
3 // Custom function banana:
4 // void = kuch return nahi
5 // blinkLED = function ka naam
6 // (int pin, int times, int waitTime) = parameters
7 void blinkLED(int pin, int times, int waitTime) {
8     for (int i = 0; i < times; i++) {
9         digitalWrite(pin, HIGH);
10        delay(waitTime);
11        digitalWrite(pin, LOW);
12        delay(waitTime);
13    }
14 }
15
16 // Value return karne wala function
17 float voltageRead(int pin) {
18     int raw = analogRead(pin);
19     float volts = raw * (5.0 / 1023.0);
20     return volts; // Value wapas bhejo
21 }
22
23 void setup() {
24     pinMode(ledPin, OUTPUT);
25     Serial.begin(9600);
26 }
27
28 void loop() {
29     blinkLED(ledPin, 3, 200); // 3 baar blink karo, 200ms gap
30
31     float v = voltageRead(A0); // Voltage opadh
32     Serial.print("Voltage: ");
33     Serial.println(v);
34
35     delay(2000);
36 }
```

Tip

Functions banane ke fayde:

- Code baar baar likhna nahi padta
- Code samajhna aasaan hota hai

- Ek jagah fix karo — sab jagah fix ho jaata hai

8 Timing — millis() ka Use

8.1 delay() vs millis()

Dhyan Do (Warning)

delay() ki problem: Jab delay() chal raha hota hai, Arduino **koi kaam nahi kar sakta** — button press bhi nahi padh sakta! Isliye bade programs mein millis() use karte hain.

Example 15: millis() se Non-Blocking Blink

```
1 int ledPin = 13;
2 bool ledState = false; // LED ki current state
3 long previousMillis = 0; // Pehli baar ka time
4 long interval = 1000; // 1000ms = 1 second
5
6 void setup() {
7     pinMode(ledPin, OUTPUT);
8 }
9
10 void loop() {
11     long currentMillis = millis(); // Abhi kitna time hua (ms mein)
12
13     // Kya interval guzar gaya?
14     if (currentMillis - previousMillis >= interval) {
15         previousMillis = currentMillis; // Time save karo
16
17         ledState = !ledState; // State toggle karo
18         digitalWrite(ledPin, ledState); // LED update karo
19     }
20
21     // Yahan aur kaam bhi kar sakte hain bina ruke!
22     // Button padhna, sensor check karna, etc.
23 }
```

Example 16: millis() se Multiple Tasks Ek Saath

```
1 int led1 = 12, led2 = 13;
2 bool state1 = false, state2 = false;
3 long prev1 = 0, prev2 = 0;
4 long interval1 = 500, interval2 = 1300; // Alag alag speeds
5
6 void setup() {
7     pinMode(led1, OUTPUT);
8     pinMode(led2, OUTPUT);
9 }
```

```
10
11 void loop() {
12     long now = millis();
13
14     // LED 1 – 500ms par blink
15     if (now - prev1 >= interval1) {
16         prev1 = now;
17         state1 = !state1;
18         digitalWrite(led1, state1);
19     }
20
21     // LED 2 – 1300ms par blink (alag speed!)
22     if (now - prev2 >= interval2) {
23         prev2 = now;
24         state2 = !state2;
25         digitalWrite(led2, state2);
26     }
27     // Dono simultaneously blink karte hain – delay() se possible
28     // nahi tha!
29 }
```

9 Sensors ke Examples

9.1 Ultrasonic Sensor — HC-SR04

Example 17: Distance Measure karna

```
1 int trigPin = 9; // Trigger pin
2 int echoPin = 10; // Echo pin
3
4 void setup() {
5     pinMode(trigPin, OUTPUT);
6     pinMode(echoPin, INPUT);
7     Serial.begin(9600);
8     Serial.println("Distance Sensor Ready");
9 }
10
11 float getDistance() {
12     // Trigger pulse bhejo
13     digitalWrite(trigPin, LOW);
14     delayMicroseconds(2);
15     digitalWrite(trigPin, HIGH);
16     delayMicroseconds(10);
17     digitalWrite(trigPin, LOW);
18
19     // Echo ka time measure karo
20     long duration = pulseIn(echoPin, HIGH);
21
22     // Distance calculate karo (cm mein)
23     // Sound ki speed = 343 m/s = 0.0343 cm/microsecond
24     float distance = duration * 0.0343 / 2;
25     return distance;
26 }
27
28 void loop() {
29     float dist = getDistance();
30
31     Serial.print("Distance: ");
32     Serial.print(dist);
33     Serial.println(" cm");
34
35     if (dist < 10) {
36         Serial.println("ALERT: Bahut paas!");
37     }
38
39     delay(500);
40 }
```

9.2 DHT11 — Temperature & Humidity Sensor

Example 18: Temp aur Humidity Read karna

```
1 // Library install karo: DHT sensor library by Adafruit
2 #include <DHT.h>
3
4 #define DHTPIN 2 // DHT11 pin 2 par
5 #define DHTTYPE DHT11 // DHT 11 ya DHT 22
6
7 DHT dht(DHTPIN, DHTTYPE); // DHT object banao
8
9 void setup() {
10     Serial.begin(9600);
11     dht.begin(); // Sensor shuru karo
12 }
13
14 void loop() {
15     delay(2000); // DHT11 ko 2 seconds chahiye
16
17     float humidity = dht.readHumidity(); // Humidity %
18     float temperature = dht.readTemperature(); // Celsius mein
19
20     // Check karo reading sahi hai?
21     if (isnan(humidity) || isnan(temperature)) {
22         Serial.println("Sensor error!");
23         return;
24     }
25
26     Serial.print("Humidity: ");
27     Serial.print(humidity);
28     Serial.print("% | Temperature: ");
29     Serial.print(temperature);
30     Serial.println(" C");
31 }
```

10 Motor Control

10.1 Servo Motor

Example 19: Servo Motor Control

```
1 #include <Servo.h> // Servo library
2
3 Servo myServo; // Servo object banao
4
5 void setup() {
6     myServo.attach(9); // Servo ko pin 9 se attach karo
7     Serial.begin(9600);
8 }
9
10 void loop() {
11     // 0 se 180 degree tak sweep karo
12     for (int angle = 0; angle <= 180; angle += 10) {
13         myServo.write(angle); // Angle set karo
14         Serial.print("Angle: ");
15         Serial.println(angle);
16         delay(100);
17     }
18
19     // 180 se 0 degree tak wapas aao
20     for (int angle = 180; angle >= 0; angle -= 10) {
21         myServo.write(angle);
22         delay(100);
23     }
24 }
```

10.2 DC Motor — L298N Motor Driver

Example 20: DC Motor Forward/Backward

```
1 // L298N Motor Driver pins
2 int in1 = 8; // Direction pin 1
3 int in2 = 7; // Direction pin 2
4 int enA = 9; // Enable pin (PWM – speed control)
5
6 void setup() {
7     pinMode(in1, OUTPUT);
8     pinMode(in2, OUTPUT);
9     pinMode(enA, OUTPUT);
10    Serial.begin(9600);
11 }
12
13 void motorForward(int speed) {
```

```
14  digitalWrite(in1, HIGH); // Aage ki taraf
15  digitalWrite(in2, LOW);
16  analogWrite(enA, speed); // Speed 0-255
17  }
18
19  void motorBackward(int speed) {
20      digitalWrite(in1, LOW); // Peechhe ki taraf
21      digitalWrite(in2, HIGH);
22      analogWrite(enA, speed);
23  }
24
25  void motorStop() {
26      digitalWrite(in1, LOW);
27      digitalWrite(in2, LOW);
28      analogWrite(enA, 0);
29  }
30
31  void loop() {
32      Serial.println("Forward...");
33      motorForward(200); // Full speed aage
34      delay(2000);
35
36      motorStop();
37      delay(500);
38
39      Serial.println("Backward...");
40      motorBackward(150); // Slow speed peeche
41      delay(2000);
42
43      motorStop();
44      delay(1000);
45  }
```


11 Interrupts — Turant Response

11.1 External Interrupt kya Hota Hai?

Normal `loop()` mein code upar se neeche chalta hai — koi cheez bich mein rok nahi sakti. Interrupt ek **emergency signal** hai jo loop ko rokta hai aur turant ek function chalata hai.

Example 21: Button Interrupt — Instant Response

```
1 volatile bool ledState = false; // volatile zaroori hai
   interrupt mein
2
3 void ledToggle() { // Yeh function interrupt par chalega
4     ledState = !ledState;
5     digitalWrite(13, ledState);
6 }
7
8 void setup() {
9     pinMode(13, OUTPUT);
10    pinMode(2, INPUT_PULLUP); // Pin 2 interrupt ke liye
11
12    // attachInterrupt(pin, function, when)
13    // FALLING = jab HIGH se LOW ho (button dabane par)
14    attachInterrupt(digitalPinToInterrupt(2), ledToggle, FALLING);
15
16    Serial.begin(9600);
17    Serial.println("Interrupt ready! Button dabao.");
18 }
19
20 void loop() {
21     // Yahan koi bhi kaam karo
22     // Button press par LED turant toggle hogi
23     // loop() ko rokna nahi padega!
24     Serial.println("Main loop chal raha hai...");
25     delay(1000);
26 }
```

Yaad Rakho (Remember)

Interrupt mein yaad rakho:

- Interrupt function mein `delay()` use mat karo
- Interrupt function chhota aur fast hona chahiye
- Shared variables ko **volatile** likhna zaroori hai
- Arduino Uno mein sirf Pin 2 aur 3 interrupt support karte hain

12 Quick Reference Card

12.1 Sabse Common Functions

Function	Parameters	Kaam
pinMode()	pin, mode	Pin ka mode set karo
digitalWrite()	pin, value	Digital output bhejo
digitalRead()	pin	Digital input padho
analogWrite()	pin, value(0-255)	PWM output bhejo
analogRead()	pin	Analog value padho (0-1023)
delay()	milliseconds	Ruko
millis()	—	Ab tak ka time (ms)
Serial.begin()	baud rate	Serial shuru karo
Serial.print()	data	Print karo
Serial.println()	data	Print karo + newline
map()	val, fL, fH, tL, tH	Range badlo
constrain()	val, min, max	Range limit karo

12.2 C++ Operators Cheat Sheet

Operator	Naam	Example
=	Assignment	x = 5
==	Equal check	if (x == 5)
!=	Not equal	if (x != 0)
> , <	Greater/Less	if (x > 10)
++	Increment	x++ (x = x+1)
--	Decrement	x-- (x = x-1)
&&	AND	if (a && b)
	OR	if (a b)
!	NOT	if (!a)
+=	Add assign	x += 5 (x = x+5)

All the Best!

Arduino seekhna ek exciting journey hai!
Chhote examples se shuru karo, experiment karo,
aur apne khud ke projects banao.

Har error ek naya seekhne ka mauka hai!